Agam Rain Cohen

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EDUCATION

The University of Edinburgh (Penultimate Year) [First Class] Sept 2023 – May 2027

Course: Computer Science (BSc Hons)

Key Modules: Algorithms and Data Structures, Data Science, Computer Systems, Software Engineering and Professional Practice

WORK EXPERIENCE

The University of Edinburgh - School of Informatics, Undergraduate Researcher Nov 2024 – Present

- Co-authored a research poster and paper on the Academic Families project, investigating the role of peer-support networks in student wellbeing and community-building within computing education.
- Applied qualitative data analysis by thematically coding and visualizing insights from 10+ interviews, generating evidence-based recommendations for enhancing student experience.
- Improved research workflows by establishing a structured process for transcribing interviews and managing participant data.
- Ensured ethical and data protection compliance by securely handling 50+ consent forms and confidential records.

The University of Edinburgh - School of Informatics, Technical Communications Intern Oct 2024 – May 2025

- Analyzed engagement metrics by programmatically collecting platform data; these insights informed a new content strategy that increased user engagement by 40%.
- Designed and executed data-driven digital campaigns for 20+ student events, implementing A/B testing on promotional materials to optimize for higher conversion rates and boost attendance.

PROJECT EXPERIENCE

MIPS Processor Simulator in C

- Developed an instruction set simulator in C capable of executing MIPS machine code programs instruction-by-instruction.
- Engineered the core fetch-decode-execute loop to process binary instructions, correctly managing the program counter, register file, and memory state.
- Implemented the logic to decode 32-bit machine code, handling a diverse subset of MIPS instructions including arithmetic, memory (load/store), control flow (jumps), and system calls for I/O operations.
- Developed a debug tracing feature to disassemble and log each instruction during execution, enabling step-by-step validation against known outputs from the MARS IDE.

Spell Checker in C and MIPS Assembly

- Developed a multi-module spell checker, first by building the complete program in C and then translating the logic statement-by-statement into MIPS assembly language.
- Engineered the core logic to load a dictionary from a file, tokenize an input text stream into individual words, and perform case-insensitive comparisons to flag misspelled words.
- Implemented fundamental low-level string manipulation routines from scratch in MIPS assembly, including strcmp, strncpy, and a custom tokenizer, while adhering to MIPS calling conventions.

FIRST Tech Challenge Robot

- Led a team to the state semifinals by architecting and programming the robot's core control system and logic in JavaScript.
- Implemented autonomous navigation routines using sensor fusion from encoders and gyroscopes, enabling the robot to score points without human intervention.
- Designed and integrated a computer vision system to detect and track field elements, increasing operational accuracy during the autonomous period.

Owl & Mouse Interactive Circuit — Two Arduino boards communicating

- Engineered an interactive, dual-microcontroller system in C++, creating a responsive display of an owl and a mouse controlled by two communicating Arduino boards.
- Implemented a non-blocking, event-driven program structure to asynchronously process inputs from light and distance sensors, enabling simultaneous actions and animations without delay.
- Developed a custom serial communication protocol to pass state changes and sensor data between the "Owl" and "Mouse" Arduinos in real-time.
- Orchestrated complex physical feedback by controlling multiple servo motors for precise movements (tongue, tail) and driving RGB LEDs for dynamic visual signaling.

LEADERSHIP

School of Informatics, Academic Parent Sept 2024 – June 2025

- Mentored a group of first-year Computer Science students, providing technical guidance on coursework and advising on effective time management and study strategies.

Robotics Competition Team, Leader Sept 2022 – June 2023

- Led the team to win over 5 regional tournaments and reach the semi-finals of the state tournament by defining technical strategy and overseeing the entire engineering lifecycle.
- Architected the robot's Javascript-based control system and mentored 5 junior members on object-oriented programming, version control (Git), and robotics principles.

SKILLS

Programming Languages: Python, Java, Bash, C, C++, Haskell, JavaScript
Tools & Technologies: GitHub, VSCode, Linux, IntelliJ, 3D Printer
Certifications: Robotics Essentials 2022 (MIT): Learned design, development, sensing, perception, decision making, motion planning, human-robot interaction, and the present and future of robotics